

Java RadioButton

Java Swing



JRadioButton

- We use the JRadioButton class to create a radio button.
- Radio button is use to select one option from multiple options.

Core Component Methods for JRadioButton

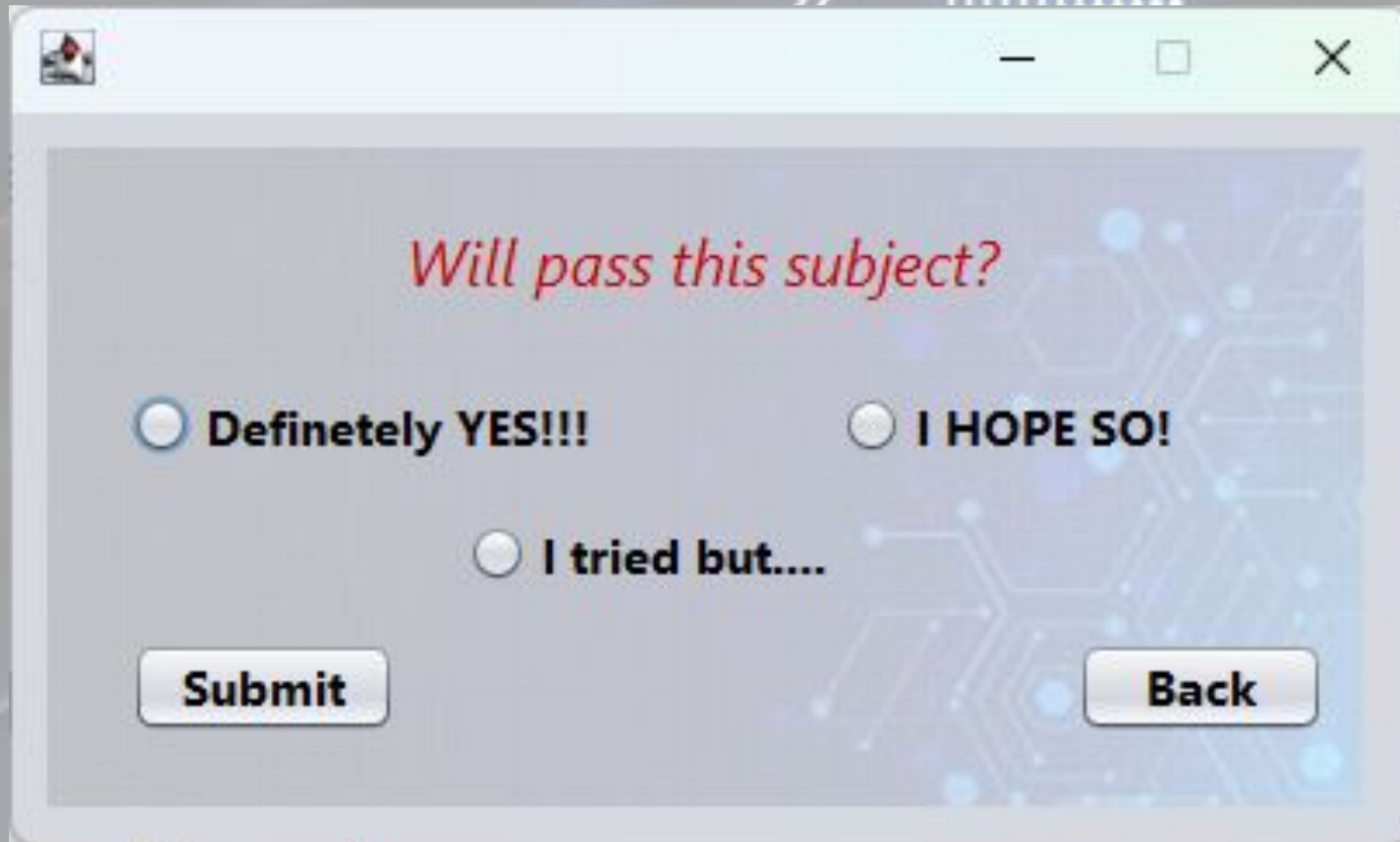
Method	Description
isSelected()	Returns a boolean (true or false) indicating if the radio button is currently selected.
setSelected(boolean b)	Sets the radio button to a selected (true) or unselected (false) state programmatically.
setText(String text)	Sets the label or caption displayed next to the radio button.
getText()	Retrieves the current text label of the radio button.
setEnabled(boolean b)	Enables or disables the radio button for user interaction.

Example

- Create a Project (FinalProject).
- Create new JFrame (You can create a Main Menu Later)
- UI needs:
 - 1 Label
 - 3 RadioButton
 - 2 Button

Java

Sample Design



Will pass this subject?

☒ **Definetely YES!!!**

☐ **I HOPE SO!**

☐ **I tried but....**

Submit **Back**

Syntax - jRadioButton

```
private void jRB1MouseClicked(java.awt.event.MouseEvent evt) {  
    if (jRB1.isSelected()) {  
        jRB2.setSelected(false);  
        jRB3.setSelected(false);  
    }  
}
```

```
private void jRB2MouseClicked(java.awt.event.MouseEvent evt) {  
    if (jRB2.isSelected()) {  
        jRB1.setSelected(false);  
        jRB3.setSelected(false);  
    }  
}
```

```
private void jRB3MouseClicked(java.awt.event.MouseEvent evt) {  
    if (jRB3.isSelected()) {  
        jRB2.setSelected(false);  
        jRB1.setSelected(false);  
    }  
}
```

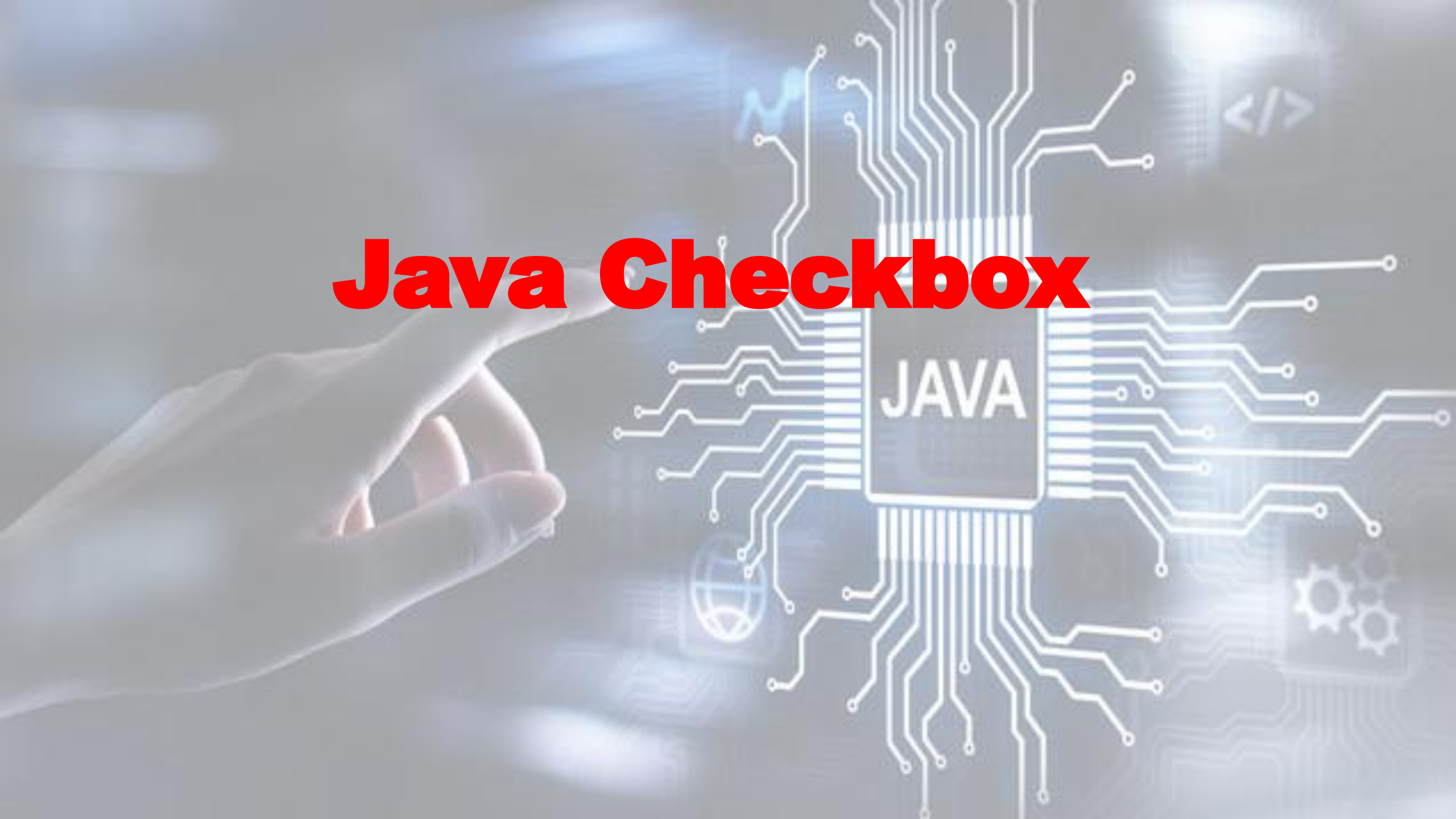

jButton1 (Submit)

```
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {  
    if (jRB1.isSelected()){  
        JOptionPane.showMessageDialog(null, "Congratulations! Your hard work and dedication have paid off");  
    } else if (jRB2.isSelected()) {  
        JOptionPane.showMessageDialog(null, "I appreciate the effort you put into this project."  
            + "Before we finalize it,"  
            + "\nI think we need to adjust something on your project!");  
    } else {  
        JOptionPane.showMessageDialog(null, "Try and Try until you succeed!");  
    }  
}
```

jButton2 (Back)

```
private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {  
    this.setVisible(false);  
    new MainMenu().setVisible(true);  
}
```


Java Checkbox



JCheckBox - Java Swing

- JCheckBox is a part of Java Swing package.
- JCheckBox can be selected or deselected.
- It displays its state to the user.
- JCheckBox is an implementation to checkbox.

Commonly used methods in JCheckBox

Method	Description
setText(String s)	sets the text of the checkbox to the given text
setIcon(Icon i)	sets the icon of the checkbox to the given icon
setSelected(boolean b)	sets the checkbox to selected if boolean value passed is true or vice versa
getIcon()	returns the image of the checkbox
getText()	returns the text of the checkbox

jCheckBox Example

- Create new JFrame

- UI needs:

- 1 CheckBox

- 1 RadioButton

- 1 Button

Java

jCheckBox Example



JRadioButton Properties

- Text: Power
- Enabled: False (uncheck)
- Font: Any (Based on your preference)
- Horizontal Alignment: Center

JCheckBox Properties

- Text: Power Switch
- Font: Any (Based on your preference)
-

Java

Syntax – Back Button

```
private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {  
    this.setVisible(false);  
    new MainMenu().setVisible(true);  
}
```

Import java awt

```
package com.mycompany.finalproject;  
  
import java.awt.Color;
```

Syntax – CheckBox

```
private void jCheckBox1ActionPerformed(java.awt.event.ActionEvent evt) {  
    if (jCheckBox1.isSelected()) {  
        jRadioButton1.setSelected(true);  
        jRadioButton1.setForeground(Color.red);  
        jRadioButton1.setText("Power is On");  
    }else{  
        jRadioButton1.setSelected(false);  
        jRadioButton1.setForeground(Color.black);  
        jRadioButton1.setText("Power is Off");  
    }  
}
```


Activity #1

JAVA



10100
01101
01010

Double VL SiLog

—

□

×

Double VL - SILOG

Price starts at 65.00

H-Silog

☐ Ham

☐ Hotdog

T-Silog

☐ Tosino

☐ Tapa

☐ Tokwa

M-Silog

☐ Maling

☐ Morcon

L-Silog

☐ Longanissa

☐ Lumpia

☐ Luncheon Meat

SubTotal:

0.00

Order

Clear

Back

Drinks

☐ SoftDrinks

☐ Coffee

☐ Hot Choco

Syntax - Order Button

```
private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) {  
    double Total;  
    Total = Double.parseDouble(jSubTotal.getText());  
    if (jCBH1.isSelected()) {  
        Total = Total + 65;  
        jSubTotal.setText(String.valueOf(Total));  
        jCBH1.setSelected(false);  
    } if (jCBH2.isSelected()) {  
        Total = Total + 65;  
        jSubTotal.setText(String.valueOf(Total));  
        jCBH2.setSelected(false);  
    } if (jRB1.isSelected()) {  
        Total = Total + 35;  
        jSubTotal.setText(String.valueOf(Total));  
        jCBT3.setSelected(false);  
    } if (jRB2.isSelected()) {  
        Total = Total + 45;  
        jSubTotal.setText(String.valueOf(Total));  
        jCBT3.setSelected(false);  
    } if (jRB3.isSelected()) {  
        Total = Total + 45;  
        jSubTotal.setText(String.valueOf(Total));  
        jCBT3.setSelected(false);  
    }  
}
```


Java List

A hand is shown on the left side, pointing towards a central graphic. The graphic features a square chip with the word 'JAVA' on it, surrounded by a complex network of white circuit lines. In the background, there are several faint, glowing icons: a blue waveform, a code symbol '</>', a globe, and two interlocking gears.

Java Swing

What is JList

- It is part of Java Swing package.
- It is a component that displays a set of Objects and allows the user to select one or more items.
- It inherits JComponent class.
- It is an easy way to display an array of Vectors .

Constructor for JList are:

- **JList()** - Creates an empty blank list
- **JList(E [] l)** - Creates an new list with the elements of the array.
- **JList(ListModel d)** - creates a new list with the specified List Model.
- **JList(Vector l)** - creates a new list with the elements of the vector

Commonly used methods are :

Methods	Description
getSelectedIndex()	returns the index of selected item of the list.
getSelectedValue()	returns the selected value of the element of the list
setSelectedIndex(int i)	sets the selected index of the list to i
setSelectionBackground(Color c)	sets the background Color of the list
setSelectionForeground(Color c)	Changes the foreground color of the list

Commonly used methods are :

Methods	Description
setListData(E [] l)	Changes the elements of the list to the elements of l .
setVisibleRowCount(int v)	Changes the visibleRowCount property
setSelectedValue(Object a, boolean s)	selects the specified object from the list.
setSelectedIndices(int[] i)	changes the selection to be the set of indices specified by the given array.

Commonly used methods are :

Methods	Description
setListData(Vector l)	constructs a read-only ListModel from a Vector specified.
setLayoutOrientation(int l)	defines the orientation of the list
setFixedCellWidth(int w)	Changes the cell width of list to the value passed as parameter.
setFixedCellHeight(int h)	Changes the cell height of the list to the value passed as parameter.
isSelectedIndex(int i)	returns true if the specified index is selected, else false.

Commonly used methods are :

Methods	Description
indexToLocation(int i)	returns the origin of the specified item in the list's coordinate system.
getToolTipText(MouseEvent e)	returns the tooltip text to be used for the given event.
getSelectedValuesList()	returns a list of all the selected items.
getSelectedIndices()	returns an array of all of the selected indices, in increasing order
getMinSelectionIndex()	returns the smallest selected cell index, or -1 if the selection is empty.

Commonly used methods are :

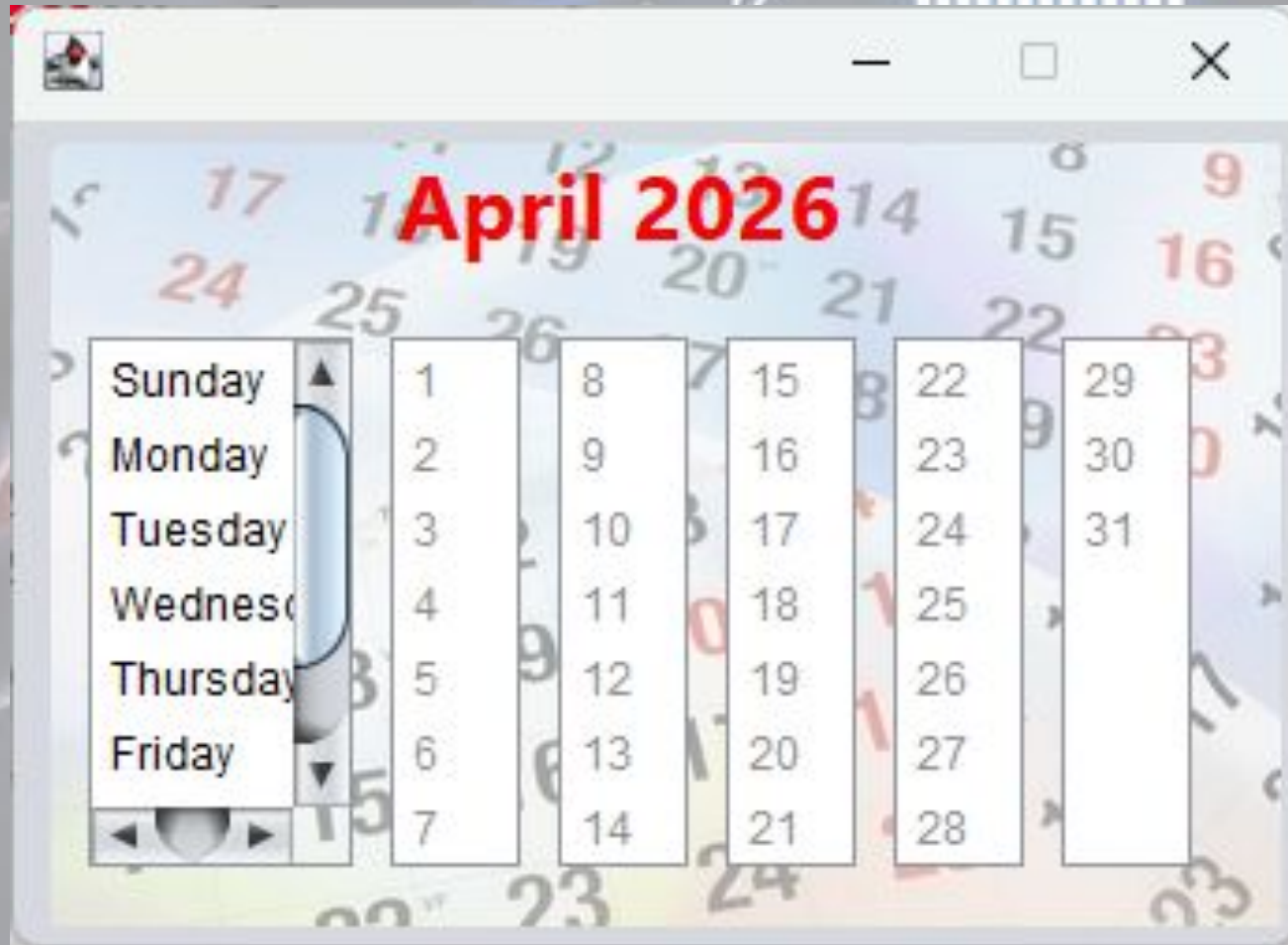
Methods	Description
getMaxSelectionIndex()	returns the largest selected cell index, or -1 if the selection is empty.
getListSelectionListeners	returns the listeners of list
getLastVisibleIndex()	returns the largest list index that is currently visible.
getDragEnabled()	returns whether or not automatic drag handling is enable

Commonly used methods are :

Methods	Description
<code>addListSelectionListener(ListSelectionListener l)</code>	adds a listSelectionlistener to the list

Java

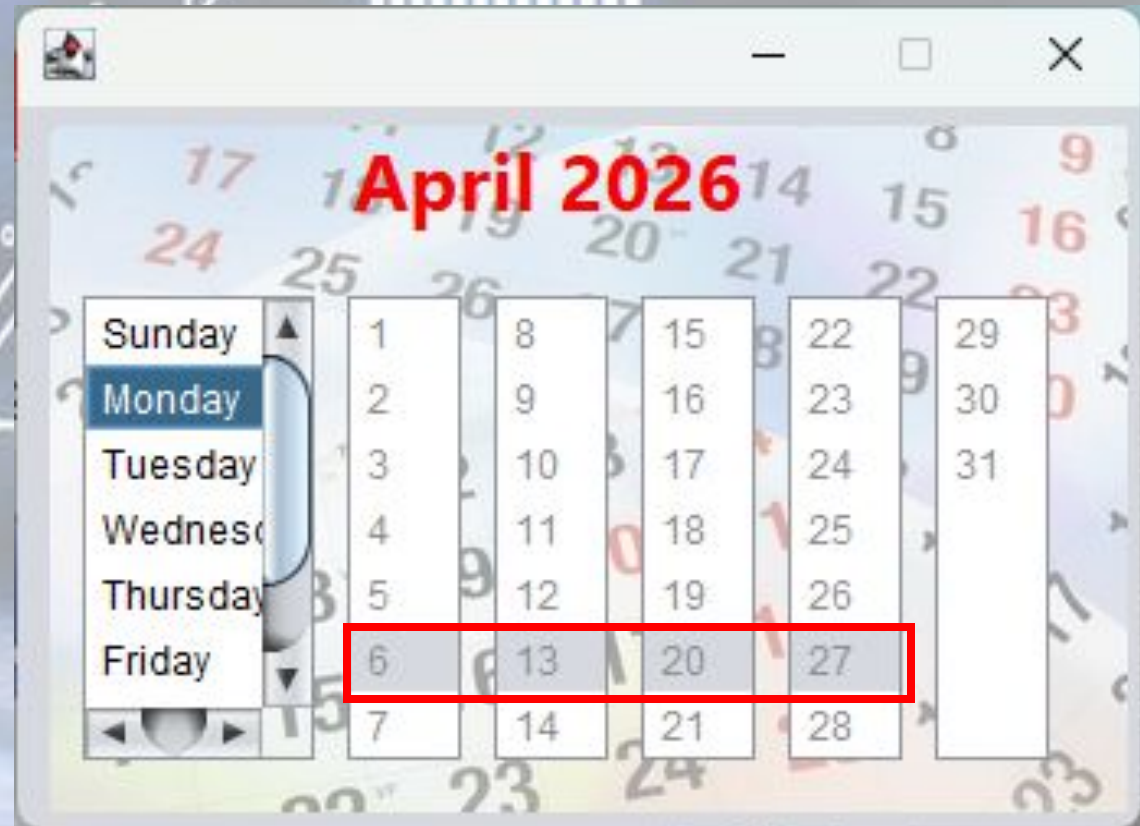
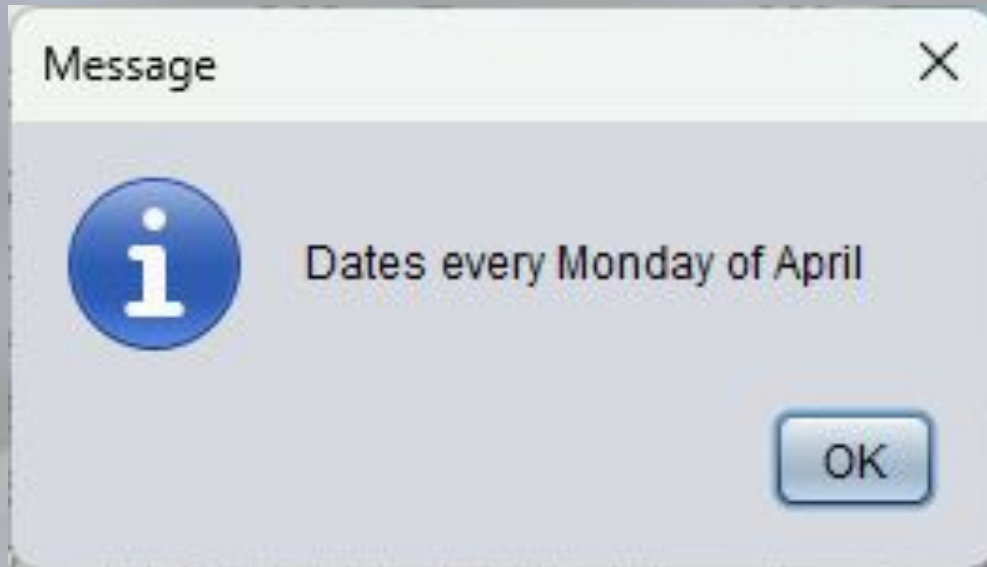
jList Example



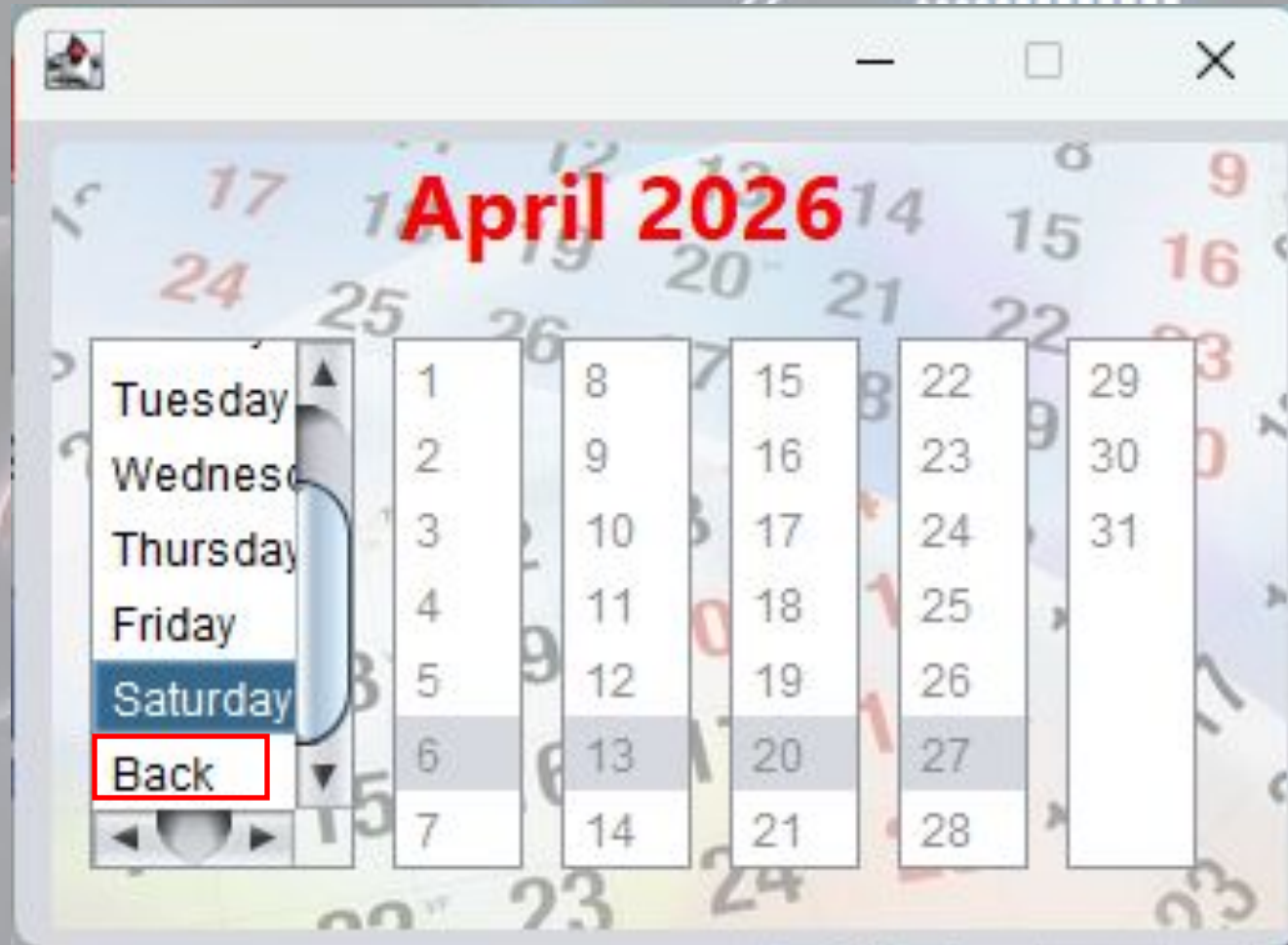
The screenshot shows a Windows calendar application. A message box is displayed in the foreground with the title "Message" and a close button (X). The message box contains an information icon (i) and the text "Dates every Sunday of April". An "OK" button is located at the bottom right of the message box. In the background, a calendar is visible, showing the month of April 2026. The days of the week are listed on the left: Sunday, Monday, Tuesday, Wednesday, Thursday, and Friday. The calendar grid shows dates from 1 to 30. The date 1 is highlighted in blue, indicating it is the current day.

A screenshot of a desktop application window titled "Calendar". The window displays a calendar for April 2026. On the left side, there is a vertical list of days of the week: Sunday, Monday, Tuesday, Wednesday, Thursday, and Friday. To the right of this list is a grid representing the calendar months. The date 5 is highlighted with a red rectangular border. The background of the calendar interface features a faint, artistic rendering of numbers in various colors (red, blue, green) scattered across it. The window has standard OS controls at the top: a minimize button, a maximize button, and a close button.

jList Example



jList Example



Syntax

```
private void jList1MouseClicked(java.awt.event.MouseEvent evt) {  
    if (jList1.isSelectedIndex(0)) {  
        JOptionPane.showMessageDialog(null, "Dates every Sunday of April");  
        jList2.setSelectedIndex(4);  
        jList3.setSelectedIndex(4);  
        jList4.setSelectedIndex(4);  
        jList5.setSelectedIndex(4);  
    } else if (jList1.isSelectedIndex(1)) {  
        JOptionPane.showMessageDialog(null, "Dates every Monday of April");  
        jList2.setSelectedIndex(5);  
        jList3.setSelectedIndex(5);  
        jList4.setSelectedIndex(5);  
        jList5.setSelectedIndex(5);  
    }  
}
```

Syntax

```
} else if (jList1.isSelectedIndex(7)) {  
    jList1.getToolTipText();  
    this.setVisible(false);  
    new Seatwork2().setVisible(true);  
}
```




Java Tabbed Pane

Java Swing

What is JTabbedPane?

- It is a GUI(Graphical User Interface) component in the Java Swing library that allows you to create a tabbed pane interface.
- It is a container that can store and organize multiple components into distinct tabs. It contains a collection of tabs.

Key Features of JTabbedPane

- **Tabs:** Each tab in a JTabbedPane is normally represented by a title or icon, and clicking on a particular tab shows the information only related to that tab.
- **Tab Placement:** JTabbedPane provides many tab placement options, such as top, bottom, left, or right sides of the pane, allowing you to customize the layout.

Key Features of JTabbedPane

- **Scrolling:** If the tabs don't fit in the visible region, JTabbedPane provides scrolling options to navigate through them.
- **Event Handling:** You can add event listeners to JTabbedPane to respond to tab selection or deselection events, allowing you to do actions when the user switches between tabs.

Constructors used in JTabbedPane

CONSTRUCTOR	DESCRIPTION
JTabbedPane()	The JTabbedPane class in Java Swing has a default, no-argument constructor called JTabbedPane(). This constructor initializes an empty tabbed pane with no tabs and no initial content when a JTabbedPane is created.

Constructors used in JTabbedPane

CONSTRUCTOR	DESCRIPTION
JTabbedPane(int tabPlacement)	The JTabbedPane(int tabPlacement) constructor allows to creation of a JTabbedPane with a defined initial location for the tabs. The tab placement option specifies whether the tabs appear at the top, bottom, left, or right of the tabbed pane.

Some commonly used methods of the JTabbedPane

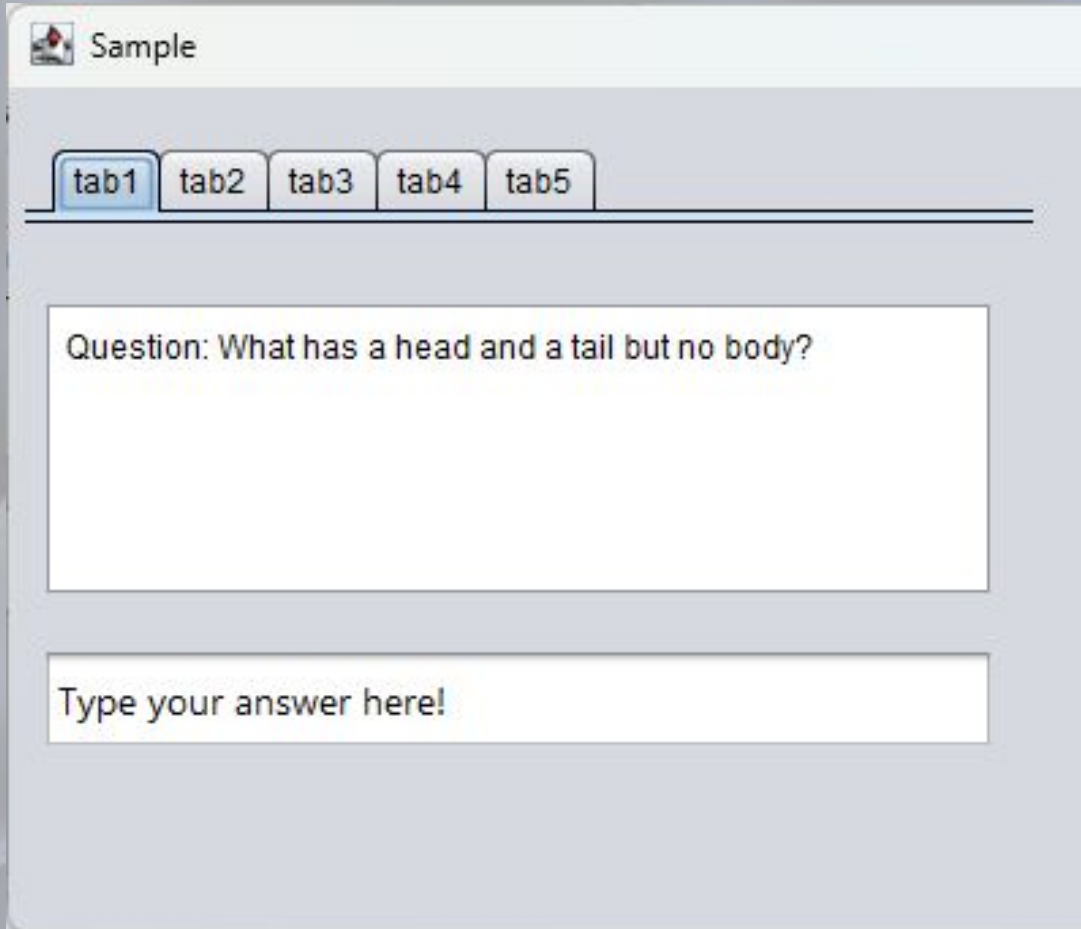
Method	Description
addTab(String title, Component component)	Creates a new tab with the given title and content.
removeTabAt(int index)	Removes the tab at the given index.
getTabCount()	Returns the number of tabs present in the JTabbedPane.

Some commonly used methods of the JTabbedPane



Method	Description
setSelectedIndex(int index)	Sets the chosen tab to the index given.
getSelectedIndex()	Returns the index of the currently selected tab.

JTabbedPane Example

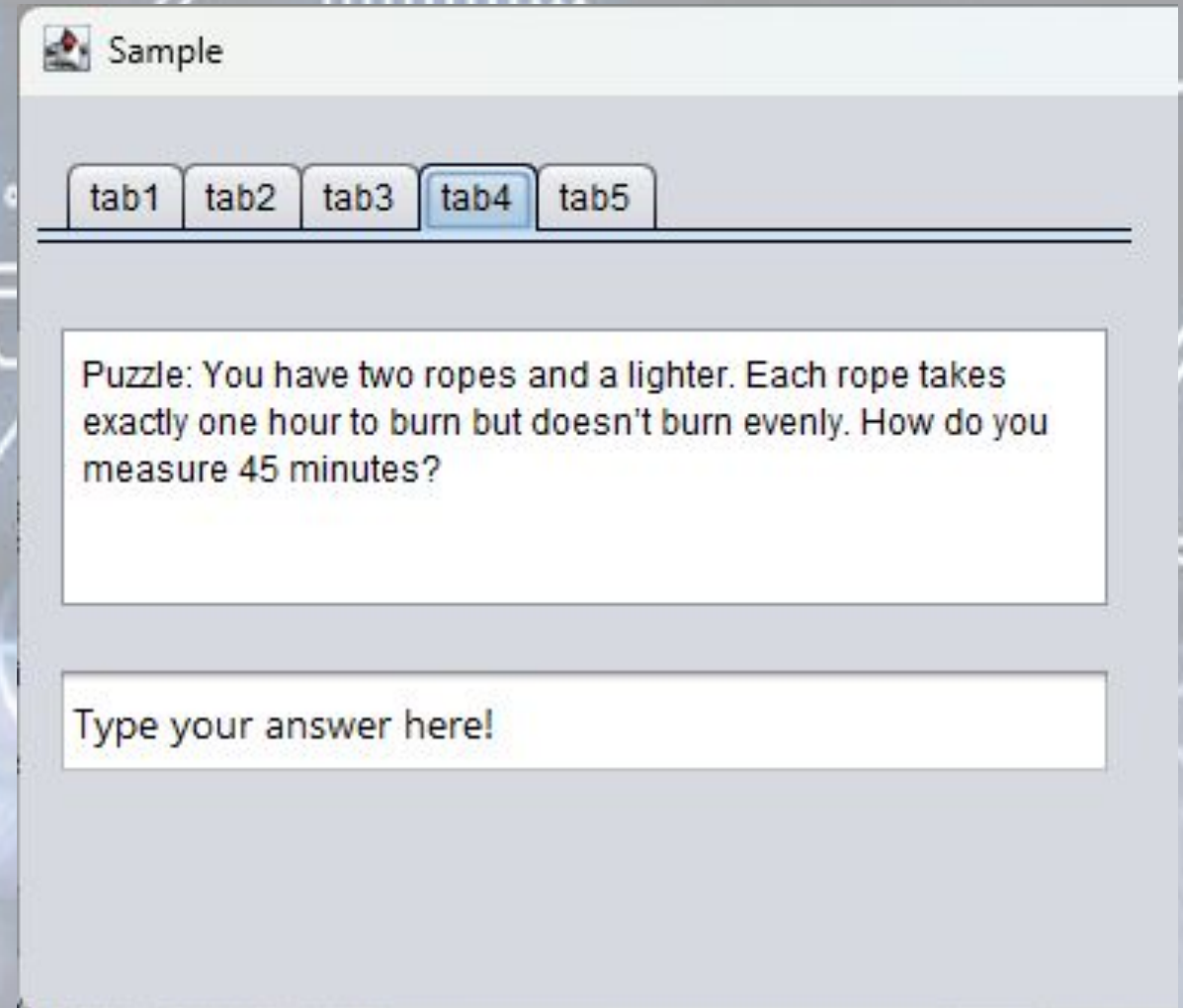
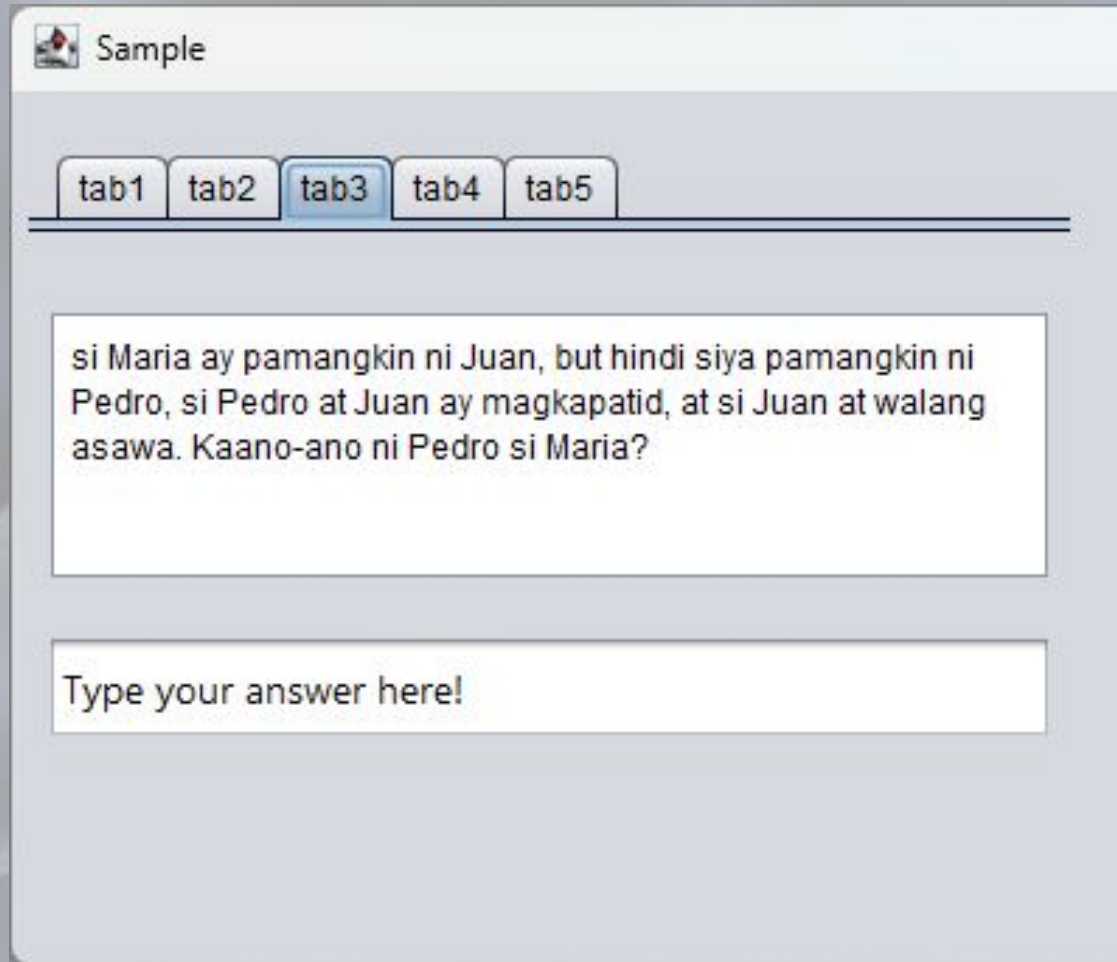


A screenshot of a Java Swing window titled "Sample". It features a JTabbedPane with five tabs labeled "tab1", "tab2", "tab3", "tab4", and "tab5". The "tab1" tab is selected and highlighted with a blue border. Below the tabs, there is a text area containing the text "Question: What has a head and a tail but no body?". At the bottom of the window, there is a text input field with the placeholder text "Type your answer here!".

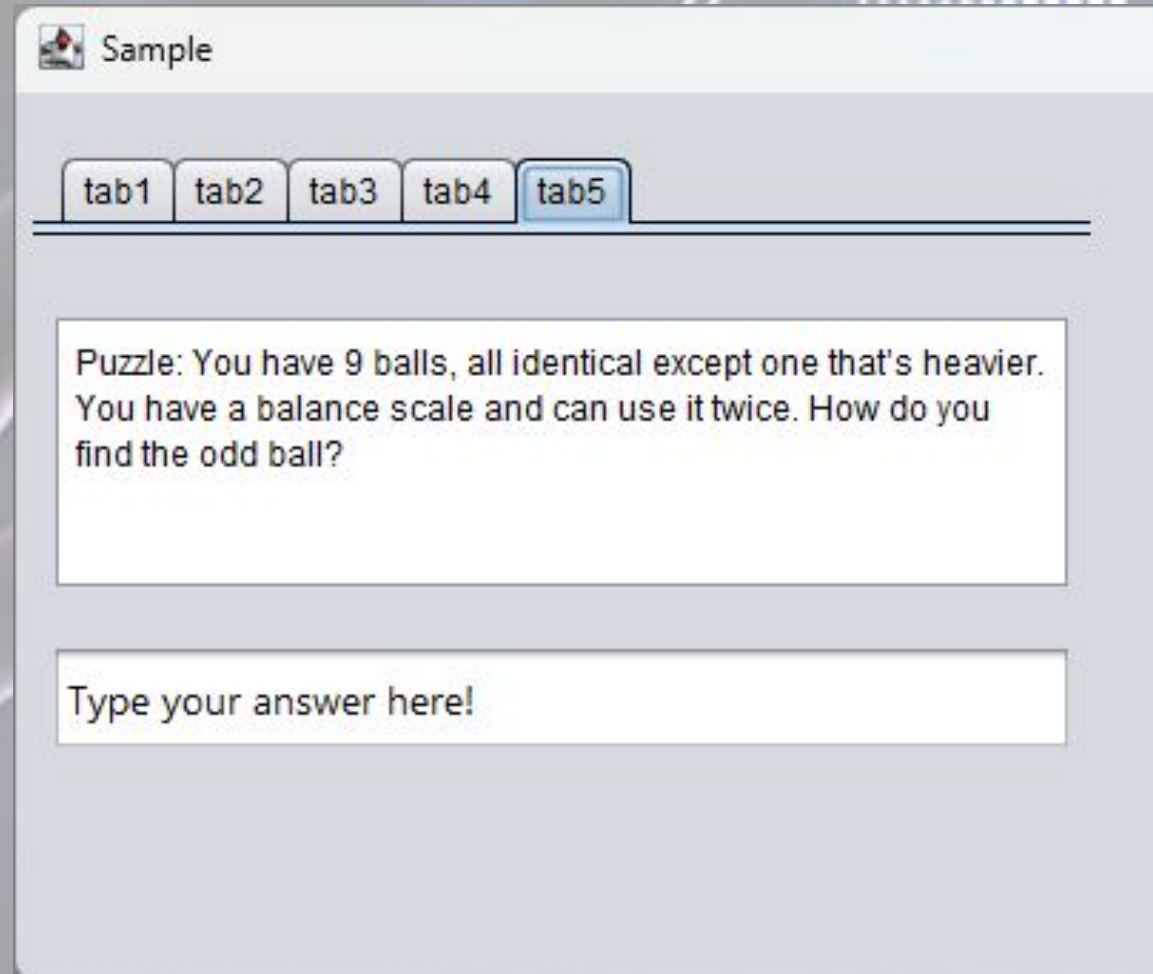


A screenshot of a Java Swing window titled "Sample", identical in layout to the first one. It features a JTabbedPane with five tabs labeled "tab1", "tab2", "tab3", "tab4", and "tab5". In this instance, the "tab2" tab is selected and highlighted with a blue border. Below the tabs, there is a text area containing the text "Why did the bicycle fall over?". At the bottom of the window, there is a text input field with the placeholder text "Type your answer here!".

JTabbedPane Example



JTabbedPane Example



Syntax

```
private void jTextField1ActionPerformed(java.awt.event.ActionEvent evt) {  
    String answer1 = jTextField1.getText();  
    if (answer1.equalsIgnoreCase("Coin")) {  
        JOptionPane.showMessageDialog(null, "Your Answer is Correct");  
        jTextField1.setEnabled(false);  
    } else {  
        JOptionPane.showMessageDialog(null, "Your Answer is incorrect.\n Try Again!");  
        jTextField1.setText("");  
    }  
}
```

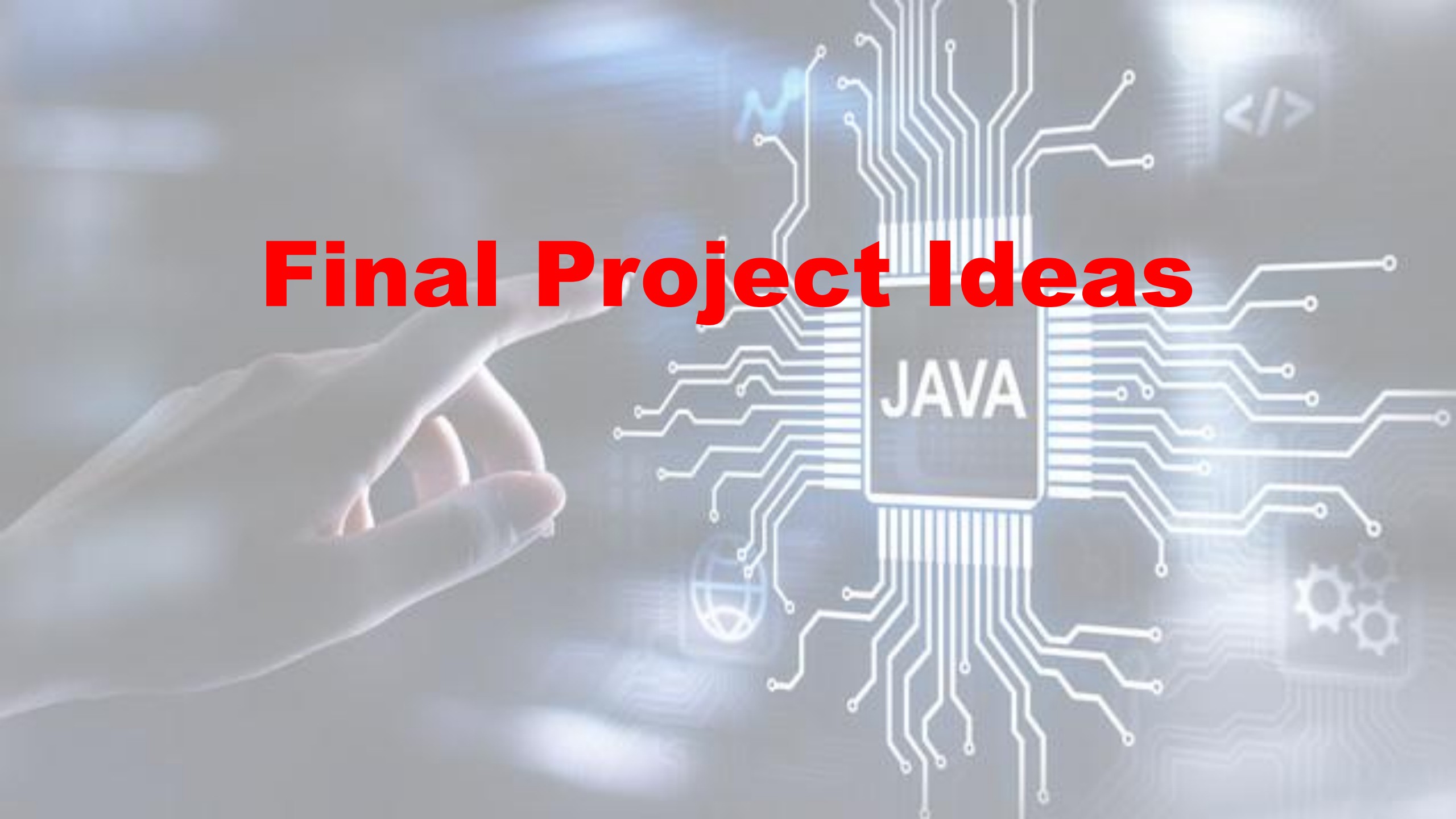

Syntax

```
private void jTextField2ActionPerformed(java.awt.event.ActionEvent evt) {  
    String answer2 = jTextField2.getText();  
    if (answer2.equalsIgnoreCase("It was too tired")) {  
        JOptionPane.showMessageDialog(null, "Your Answer is Correct\n"  
            + "it has two tires :) ");  
        jTextField2.setEnabled(false);  
    } else {  
        JOptionPane.showMessageDialog(null, "Your Answer is incorrect.\n Try Again!");  
        jTextField2.setText("");  
    }  
}
```

Syntax

```
private void jTextField3ActionPerformed(java.awt.event.ActionEvent evt) {  
    String answer3 = jTextField3.getText();  
    if (answer3.equalsIgnoreCase("Anak")) {  
        JOptionPane.showMessageDialog(null, "Your Answer is Correct");  
        jTextField3.setEnabled(false);  
    } else {  
        JOptionPane.showMessageDialog(null, "Your Answer is incorrect.\n Try Again!");  
        jTextField2.setText("");  
    }  
}
```

Final Project Ideas



Guess The Word

The image shows a software window titled "Level 1-Bill Gates Quotes". The window contains a grid of 50 empty rectangular boxes for typing a word, arranged in five rows. The first four rows have 10 boxes each, and the fifth row has 10 boxes followed by three larger, empty square boxes. Below the grid is a keyboard interface with buttons for letters A through Z. The letter 'A' is highlighted with a blue border. The window has a standard title bar with a minimize button, a maximize button, and a close button.

Level 1-Bill Gates Quotes																			

A B C D E F G H I J

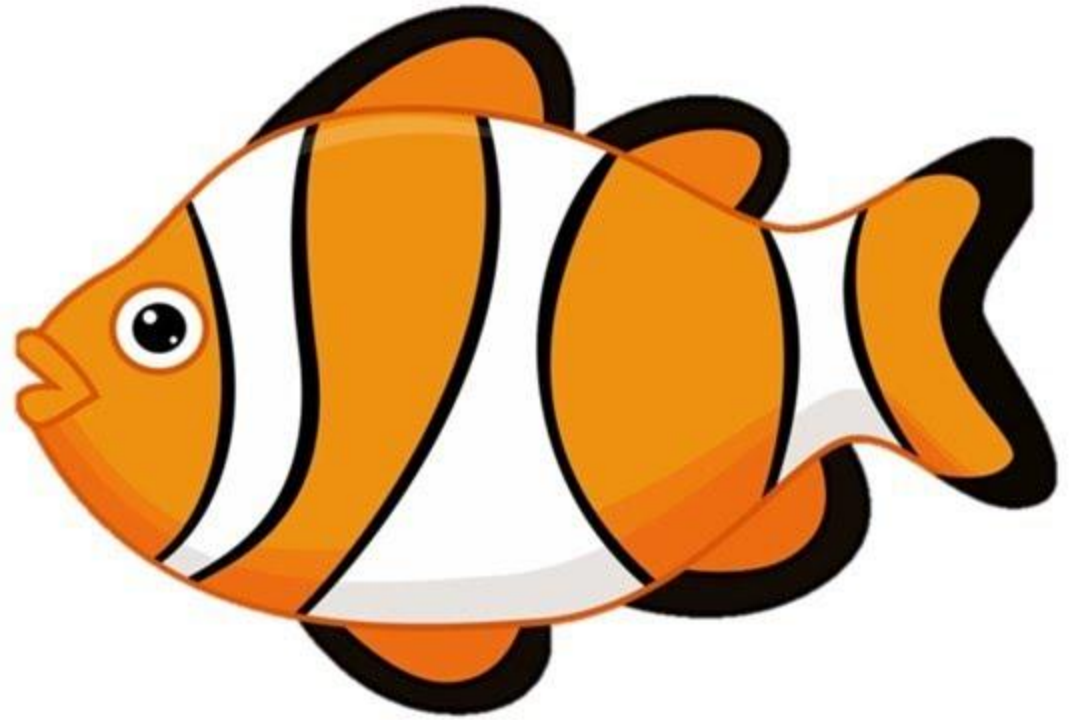
K L M N O P Q R S T

U V W X Y Z

GUESS THE WORD



+



Figerrits



Historical facts



SOLUTION

P	I	R	A	T	?	S	N	?	?	?	R	U	S	?	D
✓	✓	✓	✓	✓	6	✓	✓	6	9	6	✓	✓	✓	6	✓
T	R	?	A	S	U	R	?	M	A	P	S				
✓	✓	6	✓	✓	✓	✓	6	✓	✓	✓	✓				

You need one's __ to be their friend T R U S T
✓ ✓ ✓ ✓ ✓

He's always been her __ admirer A R D ? N T
✓ ✓ ✓ 6 ✓ ✓

Lacking any taste I N S I P I D
✓ ✓ ✓ ✓ ✓ ✓ ✓

Meant to, doomed D ? S T I N ? D
✓ 6 ✓ ✓ ✓ ✓ 6 ✓

A type of dried fruit P R U N ?
✓ ✓ ✓ ✓ 6

 6 aw in political tension D ? T ? N T ?  21
✓ 6 ✓ 6 ✓ ✓ 6

Q W E R T Y U I O P

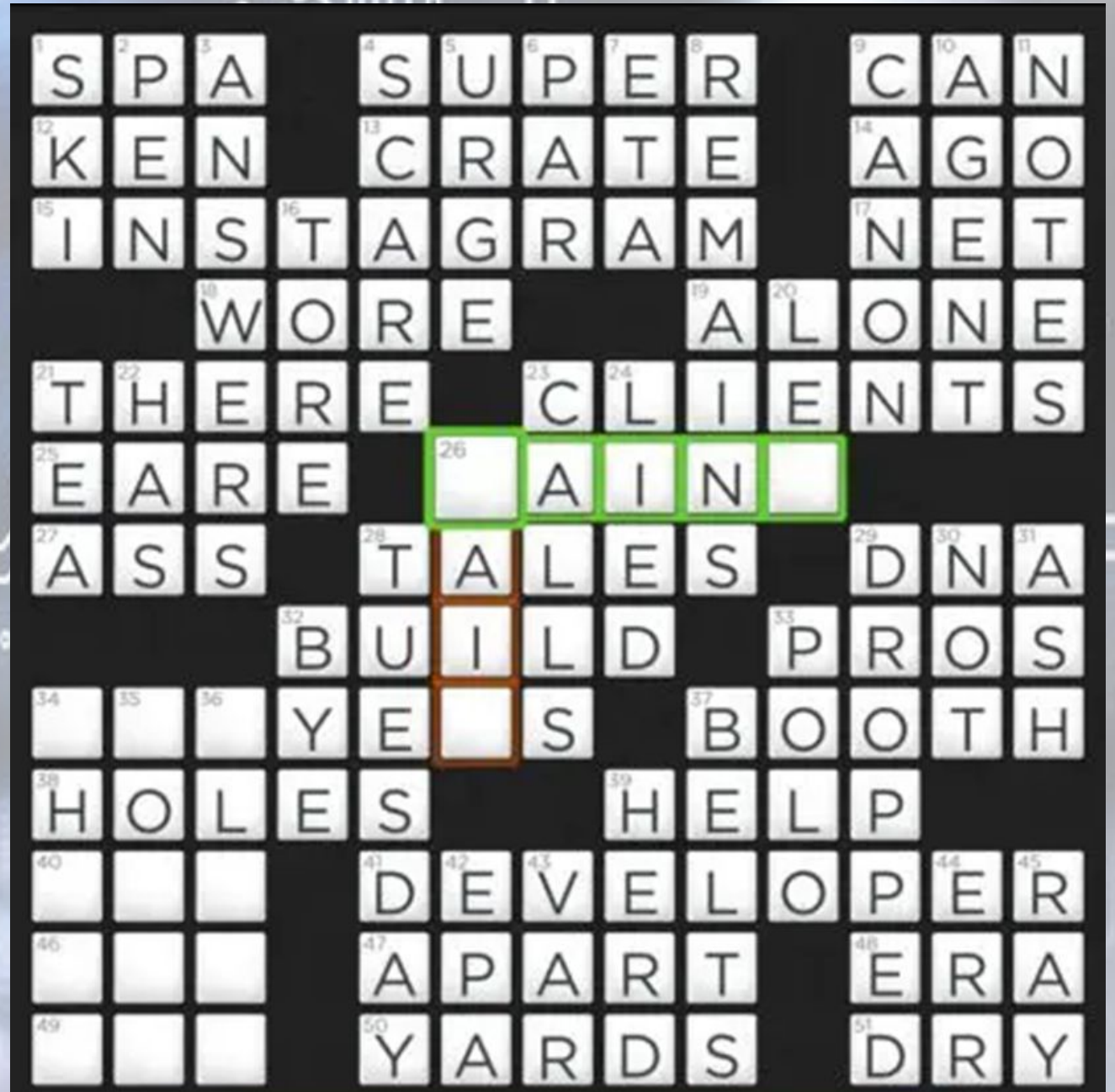
A S D F G H J K L

↩ Z X C V B N M ↵

1 word 4 pics



Crossword Puzzle





Rubrics

PROGRAM EXECUTION

Criteria	Excellent (10)	Good (7)	Poor (4)
Program Execution	Program executes correctly with no syntax or runtime errors	Program executes with a minor (easily fixed error)	Program does not execute

DESIGN LOGIC

Criteria	Excellent (10)	Good (7)	Poor (4)
Design Logic	Program is logically well designed	Program has slight logic errors that does not significantly affect the results	Program is incorrect

STANDARD

Criteria	Excellent (10)	Good (7)	Poor (4)
Standards	The program is followed on the Standard rules or Program Displayed more than Expected	The program followed some Standard rules	Program did not follow Standard rules

COOPERATION / COLABORATION

Criteria	Excellent (10)	Good (7)	Poor (4)
Cooperation/ Colaboration	Always contributes to the overall goal of the group	Usually contributes to the overall goal of the group	Rarely contributes to the overall goal of the group

RESPONSIVENESS

Criteria	Excellent (10)	Good (7)	Poor (4)
Responsiveness	The response in questions is very satisfied.	The response is quite slow	Didn't response on questions

Thank You

